What momentum says about impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect stopping impulse to average force without claiming a complete damage forecast.

SUCCESS CRITERIA - I can...

1. Compute momentum change during a stop.
2. Compare average force for different stopping times.
3. Name information beyond momentum needed for damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impact” and “momentum” to explain your model or result. Sentence starters are optional.

impact: a collision or strike between objects

momentum: mass times velocity, including direction

interval: the time or range between two stated values

01 Bring back an earlier idea

Two 3 kg carts at +5 and -3 m/s stick.
Find final velocity and use kinetic energy to explain why this is not elastic.

Word help: kinetic energy = energy associated with motion

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $|F_{avg}| = |\Delta p|/\Delta t$.
Stopping force depends on stopping time as well as momentum. These quantities alone do not specify damage.

Worked reasoning: Two incoming objects have equal momentum 12 kg·m/s but masses 2 and 4 kg, so their speeds are 6 and 3 m/s. Momentum alone does not identify every possible consequence of stopping them.

Your turn: Draw equal momentum arrows and unequal velocity arrows.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two equal-momentum objects stop in .1 s and 1 s. Which has greater average force magnitude, and by what factor?

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

04 Solve and justify

A hypothetical object of mass 10^6 kg at 200 m/s stops in .50 s.
Find impulse and average force magnitude. Then compare a 5 s stop.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why is a single stopping-force number not a prediction of buildings damaged by an asteroid?

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Choose a method

An ideal circular orbit has GM=576 m^3/s^2 and r=16 m.
Find speed, inward acceleration and period; explain how velocity changes.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

Objects A and B have equal momentum. A has twice B’s mass.
Compare speeds and kinetic energies, showing why equal momentum need not mean equal energy.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.5 kg ball arrives at +10 m/s and either stops or rebounds at -6 m/s in 0.04 s.
Compare impulse and average force.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3